



FASR Science Case # WG2-4

3D Tomography and Evolution of Quiescent and Active Region Filament Channels

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1 Science Goal(s)

The primary goal is to determine whether a filament channel is better described by a sheared arcade or by a flux rope, and to measure how that distinction changes as the channel forms, evolves, and erupts. The test uses frequency-dependent Stokes I and V morphology, spectral opacity, and the location of any gyroresonance signatures. A successful result is a ranked comparison of forward-modeled magnetic configurations with uncertainties; it is not a unique, model-free three-dimensional vector inversion.

2 Scientific Rationale

2.1 Scientific Importance

Filament channels are common sites of magnetic shear and free-energy storage. Their three-dimensional structure sets the conditions for eruption, but EUV images mainly show the cool filament and the emitting plasma around it. A sheared arcade and a flux rope can produce similar projected filament shapes while placing dense material, hot plasma, and strong field at different locations along the line of sight. Comparing their predicted radio brightness and circular polarization as a function of frequency can therefore test the magnetic topology before an eruption and follow it as it changes.

The result has direct value for CME initiation models. It can constrain which magnetic structures are present in the low corona, when axial flux becomes important, and which model assumptions fail when the channel rises. The inference must retain uncertainty in density, temperature, line-of-sight structure, and the mapping from frequency to optical depth.

2.2 Uniqueness to FASR Capabilities

EUV and X-ray observations constrain plasma temperature, density, and dynamics but do not provide the same frequency-dependent coronal magnetic diagnostic. Photospheric and chromospheric polarimetry supplies boundary conditions, with its own formation-height and opacity limitations. FASR adds spatially resolved microwave spectra and circular polarization over the channel. Frequency

is a diagnostic of opacity and, where gyroresonance is present, of field strength; it is not a direct height coordinate. The comparison must use radiative-transfer calculations and data-driven magnetic models (Alissandrakis and Gary 2021; Aschwanden et al. 2003).

2.3 Synergies

DKIST, SST, and synoptic magnetograms provide the photospheric and chromospheric boundary field. MUSE and Solar-C/EUVST can constrain density, temperature, flows, and the timing of channel formation. EUV morphology identifies the material that must be included in the radio transfer model. These observations are inputs and cross-checks for the radio inference.

2.4 Measurements Required by FASR

FASR must deliver spatially registered Stokes I and V spectra across the polarity inversion line, the filament channel, and its overlying arcade. The analysis will compare synthetic sheared-arcade and flux-rope observations after varying density, temperature, viewing angle, and unresolved structure. The two model families are distinguishable only when the correct family is recovered in source-injection tests at the adopted false-discrimination rate.

The author-submitted 1–20 GHz coverage is retained as the science-case starting interval. Its useful limits, fractional channel width, and delivered angular resolution are design-phase quantities: they must be selected by showing that they recover the morphology and opacity features that separate the models. Stokes V is a baseline product. Q and U are deferred; they become a requirement only if propagation-aware simulations show that linear polarization survives coronal depolarization and changes the model ranking.

III. Science Requirements Tables

(A) Observational Target Description

The target is one quiet-Sun or active-region filament channel and its overlying arcade. The field must include the polarity inversion line and enough surrounding corona to distinguish a local topology change from a change in background emission. Brightness and noise thresholds are source dependent. They will be set by injection tests using the modeled contrast and delivered beam; no fixed temperature target is an adopted STM requirement.

(A) OBSERVATIONAL TARGET	
Type of observation (what defines a ‘target’)	A filament channel and overlying arcade, observed in Stokes I and V over the author-submitted 1–20 GHz starting interval. The observable is frequency-dependent morphology and opacity above the polarity inversion line.
Number of targets	Baseline: one target per focused sequence. Goal: a sample at quiet-Sun, active-region, and different disk-position conditions.

(A) OBSERVATIONAL TARGET (continued)		
Size of a single target (arcsec x arcsec)	Baseline: a few arcminutes, expanded when the arcade requires it.	
Distribution of all targets (arcmin x arcmin)	Baseline: the selected channel and local active-region context.	
Peak brightness (sfu/beam or Kelvin)	Source dependent; use observed or modeled brightness distributions in the injection tests.	
RMS brightness (sfu/beam or Kelvin)	Baseline: recover the faintest model-dependent I/V morphology difference at $\geq 5\sigma$; the numerical rms is a design-phase output.	
Expected circularly polarized flux density	Stokes V (sfu/beam)	Baseline: retain signed V with calibration uncertainty small enough not to bias the model-ranking test.
	V/I	Model dependent; use predicted values including sign reversals and line-of-sight cancellation.
Expected linearly polarized flux density	Stokes Q or U (sfu/beam)	Deferred until propagation-aware simulations demonstrate recoverable information.
	Q/I or U/I	Deferred; report only simulation-tested upper limits if Q/U are produced.

(B) Spectro-Temporal Requirements Discussion

The channel can evolve over minutes to hours, while an erupting channel can change during an observing sequence. Baseline revisit cadence should therefore be shorter than the evolution time at which a static forward model ceases to be valid. A 5-minute revisit is a reasonable starting point for quasi-static evolution; a 60-second goal is appropriate during formation or eruption if it improves topology discrimination in injection tests.

The frequency grid must sample opacity changes and gyroresonance edges that differ between the model families. Use fractional spacing as the design variable. A baseline $\Delta\nu/\nu \leq 2\%$ and goal of $\leq 1\%$ should be retained only if coarser sampling changes the model ranking. The complete science spectrum must be available within the stated cadence so source evolution is not mistaken for spectral structure.

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	Author-submitted starting interval: approximately 1–20 GHz; final limits are a design-phase decision.
Instantaneous Bandwidth (GHz/pol)	Baseline: enough tunable coverage to obtain the model-discriminating spectrum within one cadence interval; implementation deferred.

(B) SPECTRAL-TEMPORAL REQUIREMENTS (continued)	
Spectral resolution [MHz]	Baseline: $\Delta\nu/\nu \leq 2\%$. Goal: $\leq 1\%$ where injection tests require it.
Temporal resolution (in seconds)	Baseline: 300 s for quasi-static evolution. Goal: 60 s during formation or eruption, subject to recovery tests.

(C) Polarization Product Discussion

Stokes I is required for thermal morphology and opacity. Stokes V is required because its sign and frequency dependence provide the principal line-of-sight magnetic constraint. Q and U are not assumed to survive propagation through the corona or unresolved field reversals. They remain optional only if propagation-aware simulations show that linear polarization survives channel averaging and changes the discrimination between sheared arcades and flux ropes.

(C) POLARIZATION DATA PRODUCTS REQUIRED	
y	Stokes I
n	Stokes Q
n	Stokes U
y	Stokes V

(D) Imaging Requirements Discussion

The spatial scale is set by the smallest morphology difference that changes the preferred topology in the forward models. The design study should test source widths and separations spanning the channel and overlying arcade, then report completeness as a function of delivered beam and largest recoverable scale. A frequency-dependent synthesized beam may be used in the forward model; a fixed common beam is not a baseline assumption.

(D) IMAGING REQUIREMENTS	
Required angular resolution (arcsec) (single value or range)	Baseline: set by source-injection recovery; design study should span approximately 5–30 arcsec.
Largest angular scale required (arcsec)	Baseline: the full filament channel and overlying arcade, determined from the target.
Mapped image size (arcmin x arcmin)	Baseline: a few arcminutes around the selected channel.
Required pixel resolution (arcsec)	Baseline: at least three pixels across the delivered beam.
Number of output/image channels	Baseline: set by fractional sampling required by topology-injection tests; no fixed channel count.

(D) IMAGING REQUIREMENTS (continued)	
Output bandwidth (minimum and maximum frequency - GHz)	Author-submitted starting interval: approximately 1–20 GHz; validate the useful subset.
Channel width (MHz)	Baseline: fractional width $\leq 2\%$; goal $\leq 1\%$ where needed.
Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each)	Baseline: $\geq 5\sigma$ recovery of the faintest I/V morphology or opacity difference that separates the models; numerical values are design-phase outputs.
Dynamic range within image (if polarization products required define for each)	Baseline: recover injected I/V signatures without a false topology preference; report the ratio from the source and calibration simulation.
Polarization accuracy (%)	Baseline: leakage and gain errors below the V signature used in the topology test; numerical accuracy is simulation-defined.
Zero spacing/total power required?	Baseline: retain diffuse brightness needed for the free-free opacity comparison; the combination with interferometric data is a design validation item.
Required maximum latency (in seconds, or N/A)	N/A for retrospective topology inference.
Required flux density scale calibration accuracy	Baseline: preserve relative spectral shape.
	GoalSet by the change in model ranking in injection tests.
	StretchAbsolute calibration sufficient for cross-instrument comparison.
	Deferred until source and transfer-model uncertainties are quantified.

2.5 Other Performance or Functional Requirements

Each snapshot must retain frequency registration, beam metadata, calibration uncertainties, observing geometry, and the sign convention for Stokes V. Those uncertainties must enter the model comparison. A target that evolves substantially during one snapshot should be modeled dynamically or excluded from the quasi-static comparison.

3 Appendix

Construct sheared-arcade and flux-rope configurations that share the observed photospheric boundary. Vary density, temperature, viewing angle, filament absorption, and unresolved structure over plausible ranges. Generate Stokes I and V data, pass them through candidate frequency sampling, beams,

calibration errors, and noise, and fit both model families. The acceptance criterion is the fraction of injections for which the correct family is preferred at the chosen false-discrimination rate. Repeat for disk position and channel evolution. Only then should numerical frequency, angular-resolution, sensitivity, and polarization requirements enter the STM baseline.

The frequency-to-height relation is model dependent. Gyroresonance can constrain field strength where the relevant harmonic is optically thick, while free-free emission constrains thermal structure and weak-field line-of-sight effects under stated assumptions. Neither diagnostic alone proves a unique three-dimensional topology. Q/U remains deferred until propagation-aware simulations show that it adds information beyond I/V.

Reference

Alissandrakis, C. E., and Gary, D. E. (2021). Radio Measurements of the Magnetic Field in the Solar Chromosphere and the Corona. *Frontiers in Astronomy and Space Sciences*, 7, 591075. doi:10.3389/fspas.2020.591075.

Aschwanden, M. J., Lee, J. K., Gary, D. E., and McTiernan, J. M. (2003). Coronal magnetic field measurements through gyroresonance emission. *ApJ*, 595, 1191–1208.